

CS 1428: Foundations of Computer Science I

Spring 2026

Section 252

Instructor: Dr. Jill Seaman
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Office Hours:

Tuesday: 9:00am–10:30am
Thursday: 1:30pm–3:00pm
Friday: 10:30am–12:00pm
in person in my office, zoom by request only.

Learning Management System: Class materials will be available on **Canvas**.

Course Format (Mode of Delivery):

This is a **Face to Face** course.

TR 3:30 pm - 4:50 pm

Centennial Hall 157

Classroom Behavior:

- Please **do not** scan in with your ID card or A# as you enter the classroom.
- Please exit the auditorium using the doors to the left of the stage.
- Please do not have discussions with your neighbor during lecture or tests.
- Please do not leave class early without permission.

Textbook:

We will be using an online textbook from **zyBooks**. It is required.

- Please see the Canvas Announcement on how to sign up for and subscribe to your zybooks book. When it asks for a student ID, **use your A0#!**
- If you opt out of BookSmart, the cost is \$89.
- For issues with zybooks: contact their support team: support@zybooks.com.

Required In-Class Response system:

We will use the **Top Hat** classroom response system in class. You will be able to submit answers to in-class questions using a mobile device or laptop. To register, go to the class Canvas site and click on Top Hat. When it asks for a student ID, **use your A0#!** Top Hat is integrated into Canvas and is free for Texas State students to use. There is a mobile app available, in addition to the Top Hat website.

Course Description:

Introductory course for computer science majors, minors and others desiring a technical introduction to computer science. The course emphasizes problem solving, algorithm development, structured programming, good coding style, and programming in C++.

Prerequisites: MATH 1315 or equivalent

Course Objectives:

At the end of the course, the students should be able to:

1. Describe the properties of good algorithms.
2. Design and develop good algorithms using a top-down approach.
3. Use the C++ programming language to implement, test, and debug algorithms for solving simple problems.
4. Explain the concepts of data types, variables, and literals and use them in programs.
5. Write C++ code that solves computational problems.
6. Use an if or if-else construct to implement branching in an algorithm.
7. Use a for loop for definite iteration.
8. Use a while or do-while loop for indefinite iteration.
9. Use functions and parameters to simplify longer programs and reuse code from previous solutions.
10. Demonstrate the mechanics of parameter passing with emphasis on the difference between pass by value and pass by reference.
11. Manipulate data in arrays.
12. Create a new data type by using a structure.
13. Analyze and explain the behavior of simple programs involving the fundamental programming constructs covered in this class.
14. Modify and expand short programs that use the constructs covered in this class.
15. Describe strategies that are useful in debugging.
16. Use a Windows- or Mac-based editor and compiler environment to develop programs in C++.

Notifications from the instructor:

Notifications related to this class will be posted in Canvas under Announcements (set up your Canvas notifications properly).

Grading:	Class Participation:	5%	Top Hat
	Participation Activities (PAs)	5%	zyBooks
	Challenge Activities (CAs)	5%	zyBooks
	Programs (MSPs)	5%	zyBooks
	Lab class:	10%	From your lab instructor
	Tests (3):	45%	Feb 19, Mar 26, Apr 23 (Thurs)
	Final Exam (comprehensive):	25%	May 12 (Tues) 2:00pm - 4:30pm

Letter Grade	Range
A	89.50 to 100%
B	79.50% to 89.49%
C	69.50% to 79.49%
D	59.50% to 69.49%
F	0% to 59.49%

Attendance policy:

Class attendance is very important but it is not directly included as part of your grade.

Class Participation:

Bring a web-enabled device to class to access Top Hat. You will be asked questions in each class (2 points each: 1 for correctness, 1 for any answer). 80% of the total possible points earns full credit for participation. You can miss up to 20% of the points and still get 100%. You may access Top Hat remotely if you are unable to attend class in person.

zyBooks:

There will be assigned readings and activities from the book each week.

PAs 10:00am Monday (every other week, starting Jan 26) (**PA**=Participation Activity)

CAs 10:00am Monday (every other week, starting Feb 2) (**CA**=Challenge Activity)

MSPs 11:59pm Monday (every other week, starting Feb 2) A portion of the MSP score will be based on our manual inspection of code, for following constraints in the instructions and style guidelines of the course. (**MSP**=Many Small Programs).

Makeup Policy:

Missed Top Hat questions and zyBooks assignments cannot be made up. The lowest of each different type of zybooks assignments is dropped.

Examinations/Tests:

- All tests will be in class on paper, using pen or pencil, during class time.
- There will be no makeup tests. If you have an instructor-approved excuse, I will replace one missing test score with your final exam score.

Course Schedule: There are 7 main topics or units:

	Topic	Dates
Unit 1	Intro to Programming & C++	Week 1 (Jan 19) Week 2 (Jan 26)
Unit 2	Expressions & Data Types	Week 3 (Feb 2) Week 4 (Feb 9)
Unit 3	Branches	Week 5 (Feb 16) Week 6 (Feb 23)
Unit 4	Loops	Week 7 (Mar 2) Week 8 (Mar 9)
Unit 5	Functions	Week 9 (Mar 23) Week 10 (Mar 30)
Unit 6	Arrays	Week 11 (Apr 6) Week 12 (Apr 13)
Unit 7	Structs	Week 13 (Apr 20) & Tues (Apr 28)

Exam coverage:

- Test 1 covers Units 1-2.
- Test 2 covers Units 3-4.
- Test 3 covers Units 5-6.
- Final Exam covers Units 1-7

Each Test/Exam will have multiple choice questions (50%) and coding questions (50%).

Withdrawals/drops:

You must follow the withdrawal and drop policy set up by the University.

<https://onestop.txst.edu/registrar/dropping-or-withdrawing.html>

Last day to drop with automatic "W": March 31, 2026.

HELP:

Other than the instructor's office hours, there are other places to obtain assistance:

- **CS Lab tutors** (good for help on MSPs): MCS 593 (inside MCS 590 suite)
<https://cs.txst.edu/resources/tutoring.html>
- **CLC tutoring** (good for help on understanding the material):
<https://hlsamp.cose.txst.edu/collaborative-learning-center.html>
- **TA's office hours** see Canvas for times and how to access.

Academic Honesty:

You are expected to adhere to the University's Academic Honor Code:

<https://www.txst.edu/honorcodecouncil/Academic-Integrity.html>

- **All assignments must be done individually**, unless the assignment instructions say otherwise.
- You may discuss problems with other students but **do not share your code electronically (or otherwise) with anyone**. Do not post your code on group chats or in email!
- **Do not include code obtained from the internet or any other source in your programs**. Do not ask Siri or use ChatGPT. Do not use features we have not covered in class in your MSPs (without prior approval of the instructor).
- **Do not collaborate with anyone during a test or exam (no talking, no cell phones or smart watches or glasses).**
- The penalty for violating academic honesty on an assignment, test or exam is a reduction of points up to a grade of 0 on the assignment, test or exam.
- If you receive a 0 for cheating on an assignment, test or exam, I will report it to the Texas State University Honor Code Council (www.txst.edu/honorcodecouncil/).

Accommodations for students with disability:

Texas State University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact Disability Services (DS) at (512) 245-3451, by email at ods@txstate.edu, or visit the [Disability Services web page](#).

Disabilities may include, but are not limited to: attentional, learning, mental health, sensory, physical, communication, or chronic medical conditions. All students are encouraged to discuss their disability related needs with the DS staff and their instructors as soon as possible.

Students who qualify for extra time for exams must take their exams with [ATSD](#) and must schedule their exam at the same time the exam is given in class. You must submit your request with ATSD at least 3 business days before the exam!

Artificial Intelligence (AI) Use in This Course

What Counts as AI

AI refers to any program or tool that can generate or edit content, including ChatGPT, Copilot, and AI features in development environments.

How You Can Use AI

You may use AI tools to support your learning, such as:

- Asking for explanations of C++ concepts or syntax
- Getting help understanding error messages
- Clarifying assignment requirements
- Learning about general programming approaches

Think of AI as a study partner. It's helpful for practice and review, but not a replacement for your own work.

Note: You are encouraged to use the AI provided by zylabs: "Help me with this test"

What's Not Allowed

- No AI-generated code: All code you turn in must be written and tested by you. Copying, adapting, or "tweaking" AI-generated code counts as prohibited use.
- No AI-submitted work: You cannot submit AI-generated writing or explanations as your own.
- No auto-complete coding from AI: If your IDE/editor has code generation features (e.g., Copilot), you need to turn them off or actively ignore them. You're responsible for ensuring your work is your own.

Why These Restrictions Exist

This course is about developing your own programming and problem-solving skills. If you rely on AI to write your code, you skip the practice you need to become a strong programmer. By writing your own solutions, you'll build skills that AI tools can't give you.

Your Responsibility

If you use AI in permitted ways, you must:

- Make sure you fully understand the explanations it provides
- Write your own original code from scratch
- Be able to explain and reproduce your solutions if asked

Misusing AI not only undermines your learning, but may also violate academic integrity policies. The goal is for you to leave this class confident in your own ability to solve problems and write programs independently.